MONASH University

Player-Pair Connections and Rotation Networks in La Trobe University Women's Basketball: UBL 2025

ETC5543 - Business analytics creative activity

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Report on La Trobe University
Women's Basketball, UBL 2025

3 November 2026

GitHub repository

github.com/mey63631/Basketball-Project

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Abstract

Coaches substitute players constantly, but individual statistics such as points and plus-minus cannot show which combinations work well together, because a player's results also depend on her teammates, her opponents and her playing time. This project examines player-pair performance and rotation structure in La Trobe University's (LTU) women's team in the 2025 University Basketball League (UBL). Event-level play-by-play data were used to reconstruct five-player lineups and same-lineup intervals ("stints") for 8 of 10 games.

A duration-weighted ridge regression (RAPM-style) estimated net points per minute from the LTU players on court and the opponent team. A pair's connection score was the duration-weighted average gap between actual and predicted performance in the stints the pair shared, reported for pairs with at least 15 minutes together across at least two games. Of 73 eligible pairs, 35 were stable positive and 28 stable negative under leave-one-game-out refits, and Player 7 appeared in 3 of the five highest-scoring pairs. Under an alternative penalty rule, rankings had Spearman correlations of 0.998 to 0.999 with the main scores. Adding pair scores did not clearly improve prediction of held-out games (duration-weighted RMSE 2.303 versus 2.300), so the scores are best used to prioritise video review, not to change rotations. Interpretation is limited by the small sample, reconstructed lineups and the use of minutes rather than possessions.

1 Background and Motivation

Coaches must decide which players to use together and when to change a lineup. Individual statistics describe aspects of player performance, but they do not establish how particular combinations perform with each other. A player's plus-minus also reflects the teammates, opponents and game situations encountered during her minutes on court (Rosenbaum 2004).

Adjusted plus-minus addresses part of this problem by regressing scoring margin on indicators for the players on court (Rosenbaum 2004). Because players repeatedly appear together, the indicators are strongly correlated and estimates can be unstable. Ridge regression penalises large coefficients to stabilise estimation (Hoerl and Kennard 1970), and Sill (2010) motivated regularisation and out-of-sample testing for adjusted plus-minus. Network methods offer a complementary view: players or teams are nodes, and centrality measures describe their position in the network (Wasserman and Faust 1994). Basketball teams have been analysed as networks of ball movement between players (Fewell et al. 2012), and eigenvector-based methods have been used to rank teams from game results (Keener 1993).

Most adjusted plus-minus work targets individual ratings. Pairs are harder, because the number of parameters grows quickly while the number of games stays small. For LTU women's basketball, an analysis of shared playing time offers a way to organise evidence about combinations and rotation patterns. Positive performance during a pair's shared minutes may indicate a combination worth investigating, but it can also reflect the contributions of the other three teammates, favourable match-ups or short-term scoring variation. Pair rankings therefore need exposure information and validation before they can inform coaching decisions.

This project was carried out in collaboration with La Trobe University. It combines an adjusted performance summary with a network description of player deployment. It separates three questions: which combinations were associated with above-prediction performance, how players were connected through shared minutes, and whether the estimated pair scores improved predictions in other games.

2 Objectives and Contribution

The primary question is: during which LTU pairings did the team perform better or worse than predicted by an additive model of its on-court players and opponent team?

The analysis has five objectives:

1. Reconstruct on-court lineups from event-level data and document the reliability of that reconstruction.
2. Calculate pair connection scores relative to an additive lineup model, reporting shared minutes and games alongside each score.
3. Describe rotation structure through a network weighted by shared playing time, and examine exposure across game situations.
4. Assess sensitivity to individual games and analyst choices, and evaluate whether a pair-score adjustment improves held-out prediction.
5. Provide league context by comparing team eigenvector centrality scores in the women's and men's competitions.

The data are play-by-play records for ten LTU games (8 retained after screening), final results for all league games, and the methods are described in Section 4. The practical contribution is a workflow for turning play-by-play records into evidence that can guide coaching review. The analytical contribution is the integration of established methods (lineup reconstruction, regularised modelling, network summaries and game-level validation) within a small single-team dataset. The project does not identify causal player chemistry or an optimal rotation.

Project contributions

- **Data collection:** a script that retrieves fixtures and play-by-play events from the competition feed, adapts to a change in the feed's response format, and saves extracted input tables for subsequent analysis.
- **Data processing:** a lineup and stint reconstruction algorithm with built-in checks, including a game-coverage screen and a comparison of reconstructed scores with the feed's scoring events.
- **Modelling and validation:** a regularised lineup model, pair connection scores, leave-one-game-out stability, sensitivity checks and a held-out prediction test.
- **Networks and visualisation:** shared-minute networks with centrality and concentration measures, and anonymised figures for reporting.
- **Reproducibility:** analysis scripts that run in a fixed order and save the outputs this report reads (Appendix A).

Skills developed and challenges

Data work. A substantial part of the project involved turning event records into an analysable dataset. I worked with nested JSON responses, adapted the extraction function when the feed's response structure changed, and tracked substitutions occurring at identical timestamps. These tasks required careful attention to event ordering and to the distinction between player identifiers and display names.

Difficulties and how they were addressed. Three difficulties shaped the analysis. First, in two games the inferred roster drifted away from five players after the first substitutions; I traced the roster size after each substitution, identified the pattern and excluded games below a coverage threshold, reporting the exclusion. Second, a data-structure error silently emptied the opponent lineups; it was found by checking lineup lengths and corrected. Third, the live feed changed during the project, so the inputs were saved as fixed snapshots and the analysis was separated from report rendering.

Analytical learning. The reconstruction checks changed how I interpreted data quality: matching scoring totals is reassuring, but it does not show that the inferred players are correct. The modelling stage developed my understanding of regularisation, game-level cross-validation and weighted-network measures. An important lesson was that a descriptive pattern can remain visible without clearly improving held-out predictions, and that such a result should be reported rather than hidden.

Workload. The work moved through stages that were each checked before the next: building and testing the extraction function; reconstructing and checking lineups against scoring events; fitting and validating the model; calculating network measures; and organising the code into scripts with a single runner, saved inputs and documentation so that results can be regenerated.

3 Data and Preparation

Source and scope

The data concern LTU women's games in the 2025 UBL. Fixture and play-by-play records were retrieved from the Sportradar feed embedded in the UniSport UBL website. Fields include fixture identifiers, period, game clock, team, player identifier, event type, event subtype, substitution direction and scoring outcome. Persistent player identifiers are used for processing. To protect privacy, players are reported as Player 1 to Player 16 throughout.

The saved play-by-play input contains 10 LTU games and 5,936 events. Of these games, 8 were retained after excluding games with reconstructed LTU playing-time coverage below 50% of an assumed 40-minute game. Among the retained games, 6 had complete reconstructed coverage and matching scoring totals. The other 2 games had coverage ranging from 79.0% to 84.5%. Retention therefore does not establish complete coverage or correct lineup identities. Table 1 summarises the coverage and scoring checks for the retained games.

Of the 10 games, 2 were excluded because lineup tracking became unreliable early in the game: after the first substitutions, the inferred roster drifted away from five players, which is consistent with an incorrect inferred opening lineup. Reconstructed LTU coverage in these games was 13% to 13% of regulation time, below the 50% threshold. The checks cannot show whether the underlying event records were incomplete.

Table 1: *Reconstruction checks for retained LTU games. Coverage is reconstructed playing time as a percentage of an assumed 40-minute game. Point totals are shown as LTU–opponent.*

Game	Coverage (%)	Reconstructed points	Feed points	Check
Game 1 vs VU	100.0	84–62	84–62	Match
Game 2 vs AU Waatu	84.5	79–60	89–72	Investigate
Game 3 vs MELB	100.0	75–57	75–57	Match
Game 4 vs UON	79.0	55–47	68–64	Investigate
Game 5 vs CU	100.0	74–72	74–72	Match
Game 6 vs UWA Maali	100.0	82–60	82–60	Match
Game 7 vs UniSC Mari	100.0	93–83	93–83	Match
Game 8 vs MELB	100.0	62–99	62–99	Match

Lineup reconstruction

Within each game, events are ordered by period and descending game clock, preserving source order for tied timestamps. Opening lineups are inferred from distinct players with recorded actions before the team's first substitution. Where five starters cannot be identified, lineup tracking may fail or be incomplete.

Substitutions update the inferred on-court roster. Consecutive substitution events at the same timestamp are processed together, and stints are closed at substitution times and period boundaries. A stint is a continuous interval during which the inferred lineup is unchanged. Points scored and conceded (made two-point shots, three-point shots and free throws) are accumulated within each stint.

The final lineup of a period is carried into the next period. The implementation checks for activity inconsistent with that lineup but flags discrepancies rather than correcting them. This is an assumption of the reconstruction procedure: players can change between periods without an in-period substitution record. The procedure flagged no quarter-break lineup discrepancies in the saved-input analysis. This does not independently verify lineup identities.

Stints shorter than 15 seconds are excluded from regression and connection scoring to reduce extreme rate estimates. The rotation-network calculations use a separate sample of all positive-duration stints, so the two samples differ in total minutes and pair counts.

Algorithm 1: Stint reconstruction and screening

1. For each fixture, order events by period, descending game clock and original source order.
2. Infer each team's opening lineup from distinct players with recorded actions before its first substitution.
3. At a substitution event, close the current stint, then process consecutive substitution records sharing the same period and timestamp in source order.
4. Accumulate points from made two-point shots, three-point shots and free throws within the current stint.
5. At each period boundary, close the stint and carry the final lineup forward, flagging activity inconsistent with that lineup.
6. Record stint duration, both inferred lineups, points scored and conceded, and the score margin at stint start.
7. Exclude games below 50% reconstructed LTU playing-time coverage.
8. For regression and connection scoring, retain stints lasting at least 15 seconds.

Data provenance and quality

All results in this report use the saved input snapshot `data/raw/LTU_women_inputs_2025.rds`, containing 5,936 play-by-play events across 10 LTU games. The snapshot records its export

time as 2026-11-01 14:03:52.80575; this is not necessarily the original collection date. Rendering the report does not download new data. The extraction function was adapted to handle different API response structures encountered during the project, and because the live feed may change, the saved inputs keep the analysis consistent across reruns.

Reconstruction checks compare accumulated stint scores with totals calculated from scoring events in the same feed. Scoring totals matched in 6 of 8 retained games (Table 1), and the checks recorded 0 negative-duration stints. In 2 retained games, coverage was incomplete or scoring totals differed from the feed, which may affect player and pair estimates.

These checks assess internal consistency rather than accuracy against an independent source. Complete playing-time coverage and matching scores do not verify every lineup: an inferred lineup can contain five players while identifying the wrong five. The checks establish no independent validation against official scores or video.

4 Methodology

Response and additive lineup model

For stint s, net points per minute is

$$y_s = \frac{P_{s,\mathrm{LTU}} - P_{s,\mathrm{opp}}}{d_s/60}, \tag{1}$$

where P denotes points scored in the stint and d_s is stint duration in seconds. The additive model is

$$y_s = \beta_0 + \sum_p \beta_p x_{sp} + \sum_o \delta_o z_{so} + \varepsilon_s, \tag{2}$$

where $x_{sp} = 1$ if LTU player p is on court and $z_{so} = 1$ if the opponent is team o. The model uses duration weights, so longer stints contribute more to fitting. The fitted prediction for stint s is denoted $\hat{y}_s$.

Ridge regression is used because the player indicators are strongly related: players repeatedly appear together, and each valid lineup contains five players. The model is fitted with the `glmnet` package (Friedman, Hastie, and Tibshirani 2010) using $\alpha = 0$, which minimises (up to the scaling used by `glmnet`)

$$\sum_s d_s \Big(y_s - \beta_0 - \sum_p \beta_p x_{sp} - \sum_o \delta_o z_{so} \Big)^2 + \lambda \Big(\sum_p \beta_p^2 + \sum_o \delta_o^2 \Big). \tag{3}$$

The implementation uses `glmnet`'s default predictor standardisation, so the ridge penalty operates on coefficients for standardised predictors. The equation presents the model's structure; the coefficients returned by the software are expressed on the original predictor scale.

Entire games are assigned to cross-validation folds, and the penalty λ is the largest value whose cross-validation error is within one standard error of the minimum (`lambda.1se`).

This is a single-team RAPM-style model, not a full model of players on both teams. The opponent indicators identify the opposing team but not which opposing players were on court. The code penalises the opponent coefficients along with the player coefficients, so they are regularised opponent-team controls rather than unpenalised fixed effects.

Because the player indicators sum to five and the opponent indicators sum to one in every stint, the unpenalised design is redundant with the intercept. Ridge regularisation still yields a fitted solution, but individual coefficients depend on the penalty and coding. They should not be read as isolated causal contributions of individual players.

Pair connection scores

The fitted model gives a residual $y_s - \hat{y}_s$ for each stint. For players i and j, the connection score is

$$C_{ij} = \frac{\sum_{s:\, i,j \in L_s} d_s \, (y_s - \hat{y}_s)}{\sum_{s:\, i,j \in L_s} d_s}, \tag{4}$$

where L_s is the inferred LTU lineup in stint s. A positive score means LTU performed above the additive lineup prediction during that pair's shared stints. Its unit is net points per minute above prediction.

The prediction behind each residual includes the intercept, all five LTU player indicators and the opponent indicator. The score is therefore not raw pair performance minus two player ratings. It is also not a separately estimated pair-interaction coefficient: each five-player stint contributes the same residual to ten overlapping pairs. The residuals are in-sample.

Pairs are reported when they share at least 15 model-eligible minutes across at least two games. These thresholds reduce reliance on brief appearances but do not guarantee precise estimates, so each score is reported with minutes, games and a stability summary.

Interpreting a connection score

A hypothetical example illustrates how the score is calculated. Suppose LTU outscores its opponent by four points during a two-minute stint. Observed net performance is therefore two points per minute. If the additive model predicts one net point per minute for that lineup and opponent, the stint residual is positive one net point per minute. This residual contributes to the connection score of each of the ten pairs in the five-player lineup.

Suppose one of those pairs also shares a four-minute stint with a residual of negative 0.5 net points per minute. Across these two stints, its duration-weighted residual is zero: the two-minute positive contribution is offset by the longer negative contribution. These hypothetical six shared minutes would not themselves meet the study's 15-minute eligibility threshold.

The example shows why connection scores depend on both performance and exposure. It also shows why they cannot isolate chemistry between two players: the same residual is attributed descriptively to several overlapping pairs, while the other teammates and game circumstances also contribute to the outcome.

Pair scores and player-level residuals

As an exploratory diagnostic, each player's duration-weighted mean stint residual was calculated across her model-eligible stints. For each eligible pair, the two player-level means

were averaged and compared with the pair connection score using Spearman rank correlation. Both quantities are derived from the same stint residuals, so this comparison describes the overlap between the summaries rather than independently identifying individual or pair effects.

Leave-one-game-out sign stability

Each retained game is removed in turn, the model is refitted, and eligible pair scores are recalculated. A pair is classified as stable positive if its score is positive in at least 80% of available refits, and stable negative if that proportion is at most 20%. Other pairs are classified as uncertain. Pairs with fewer than half as many estimates as retained games are labelled as having insufficient stability evidence; with eight games, at least four valid estimates are needed.

These labels describe sensitivity to omitting a game. They are not confidence intervals or significance tests, and they are not evidence of future predictive performance. Each refit reuses most of the same games, so the labels are far from independent validation.

Sensitivity to penalty selection and exposure thresholds

The main analysis uses `lambda.1se` and requires 15 shared minutes across at least two games. Alternative specifications used `lambda.min` and shared-time thresholds of 10 and 20 minutes, retaining the two-game requirement. Comparisons with the main analysis used pairs eligible under both specifications. Spearman correlation summarised agreement in ranking, and sign agreement measured the proportion of pairs retaining the same positive or negative sign.

The exposure threshold determines which pairs are reported. Under a fixed penalty rule, changing this threshold does not change the scores of the pairs that remain eligible, so agreement across thresholds is expected by construction. Changing the penalty rule provides a separate check because it changes the fitted baseline and the residuals.

Rotation networks and game situations

A separate undirected usage network has players as nodes and shared playing time as edges. A pair enters the network after at least one shared minute in at least one game, unlike the stricter connection-score threshold. Weighted strength is the sum of a player's edge weights. Weighted betweenness is calculated using inverse shared minutes as network distances, so frequently used pairings are treated as shorter connections (Freeman 1978; Csárdi and Nepusz 2006).

High betweenness indicates a position on shortest paths in this shared-minute network. It does not measure ball movement, leadership, passing skill or causal importance. Weighted

strength is closely tied to exposure, since each minute played adds roughly four shared-player minutes. Fewell et al. (2012) examined basketball teams as networks of ball movement between players; this project uses a different relationship, shared court time, so its centrality measures describe rotation structure rather than passing patterns.

Concentration is summarised with the Gini coefficient, where 0 means equal and 1 means fully concentrated. Freeman's betweenness centralisation is computed on the unweighted network, because its normalising constant is derived for unweighted graphs (Freeman 1978).

Game situations use the score margin at stint start: leading by at least six, within five points, or trailing by at least six. A clutch situation is a stint starting in the final five minutes of the fourth period with an absolute margin of at most five. A stint that crosses a threshold is not split.

Held-out prediction

In each outer evaluation fold, one game is held out. The baseline model, including penalty selection, is fitted on the other games, and pair scores are also computed from those games. For each test lineup, the ten pair scores are averaged (missing scores count as zero) and added to the baseline prediction. This is a fixed-coefficient heuristic, not a fitted interaction model.

Predictions are compared using duration-weighted root mean squared error,

$$\mathrm{WRMSE} = \sqrt{\frac{\sum_s d_s (y_s - \hat{y}_s)^2}{\sum_s d_s}}, \tag{5}$$

where $\hat{y}_s$ is the held-out prediction from the model being evaluated.

As an exploratory benchmark, the held-out pair feature is shuffled across pooled stints 2,000 times, with a fixed random seed, and the error improvement is recomputed. The reported p-value is the share of shuffled improvements at least as large as the observed one. Pooled shuffling ignores dependence within games, so this p-value is indicative rather than definitive.

Test players and opponents not seen in training receive no coefficient. Most opponents were met in only one retained game, so for those games the opponent term is unavailable and the baseline relies on the players and the intercept.

Interpreting the validation checks

The analysis uses several checks that address different sources of uncertainty. Reconstruction diagnostics assess whether inferred playing time and accumulated scores are internally

consistent with the event feed. They help identify missing coverage and scoring discrepancies, but cannot establish whether every player was correctly identified.

Leave-one-game-out sign stability and the sensitivity checks assess how much the descriptive results depend on individual games and analyst choices. A consistent sign suggests that a result is not entirely dependent on a single game, but the refits share most of their observations, and consistency does not establish predictive usefulness.

Held-out prediction asks a separate question: whether the specified adjustment helps predict an omitted game. That game's outcomes are excluded from baseline fitting, penalty selection and pair-score estimation. The overall weighted RMSE pools the held-out stint errors using their durations; it is not the arithmetic mean of the game-level RMSE values. Together, these checks distinguish internal consistency, sensitivity to observed games and analyst choices, and performance beyond the estimation sample.

League-level results networks

The league comparison examines a different level of basketball organisation from the LTU player analysis. In the player networks, nodes represent players and edges describe shared court time or above-prediction performance. In the league networks, nodes represent university teams and edges describe match results. The purpose is to place the LTU analysis within a broader description of the recorded competition, rather than to validate the player-pair scores. Women's and men's results are analysed separately because they concern different teams, players and fixtures.

Each network uses eligible confirmed results with recorded scores from the saved input snapshots. The team-level analysis does not require reconstructed player lineups. Consequently, excluding an LTU game because of insufficient play-by-play reconstruction coverage does not remove its result from the league comparison: a game can have a usable final result even when its event records do not support reliable lineup reconstruction.

Constructing the season network

Each participating team is represented by one node. Every eligible match contributes a directed edge from the losing team to the winning team, with weight equal to the absolute difference between their scores. For example, if Team A defeats Team B 75–65, the match contributes an edge from B to A with weight 10. This is a hypothetical example illustrating the construction, not a result from the dataset.

The direction allows winning teams to receive incoming connections from the teams they defeated. Winning margin determines the strength of each connection, so a larger victory contributes more weight than a narrow victory. This choice uses information beyond the binary win–loss outcome, but it also gives unusually large margins substantial influence.

The implementation retains a separate edge for each match, so repeated results between the same teams each contribute their own weight. If two teams defeat each other in dif-

ferent matches, connections occur in both directions. The resulting network combines all eligible results in the snapshot into one aggregate representation for each competition.

This construction describes the recorded season retrospectively. It does not update rankings sequentially after each round or evaluate whether rankings available at an earlier date predict subsequent results. It also does not adjust winning margins for venue, player availability or the number of possessions.

Eigenvector centrality and concentration

Eigenvector centrality gives each team a score that depends on the scores of the teams pointing towards it. With edges directed from losers to winners, a team receives weighted contributions from the opponents it defeated. Those opponents' centralities depend recursively on their own results. The measure therefore summarises a team's position within the wider pattern of recorded wins and margins, rather than considering its victories independently (Bonacich 1972; Keener 1993).

Let A_{ji} denote the sum of winning margins from matches in which team i defeated team j. The centrality score c_i satisfies

$$\lambda c_i = \sum_j A_{ji} c_j, \tag{6}$$

where λ is the corresponding eigenvalue. Scores are scaled so that the highest-scoring team in each competition has a value of one. This scaling supports comparison of relative positions within a competition. A score of 0.5 should not be interpreted as half the playing ability of the leading team, and scores are not official ladder positions or estimated win probabilities.

The Gini coefficient summarises inequality in these centrality scores. A larger Gini indicates that the computed scores are distributed more unevenly across teams. Its interpretation depends on the network construction: winning margins, repeated meetings, schedule differences and connectivity can all influence the distribution. The comparison is therefore a descriptive summary of results-network concentration, rather than a complete assessment of competitive balance.

Software is listed in Appendix A.

5 Results

Pair performance and stability

Table 2 and Figure 1 show the highest and lowest connection scores. Of the 73 eligible pairs, 35 were stable positive, 28 stable negative, 10 uncertain and 0 had insufficient evidence. Scores ranged from -1.45 to 1.12 net points per minute relative to prediction. Player 7 appears in 3 of the five highest-scoring pairs. The model's individual ratings were small, ranging from -0.000 to 0.000, which is consistent with strong shrinkage.

The extremes are selected from many pairs, so they probably overstate any real effect. The stability labels also reuse the same games, so they show sensitivity to single games rather than evidence of repeatability.

Table 2: *The five highest and five lowest pair connection scores. Actual and predicted are net points per minute during shared stints; score is their difference. Refits is the number of leave-one-game-out refits in which the pair was eligible.*

Pair	Min	Games	Actual	Pred.	Score	Stability	Refits
Player 16 + Player 7	24.97	3	1.28	0.17	1.12	Stable +	7
Player 6 + Player 7	17.02	3	1.23	0.17	1.07	Stable +	5
Player 10 + Player 12	15.22	2	1.18	0.17	1.02	Stable +	6
Player 12 + Player 1	15.17	2	0.92	0.17	0.76	Stable +	6
Player 4 + Player 7	59.38	4	0.89	0.17	0.73	Stable +	8
Player 13 + Player 14	15.57	4	-1.28	0.17	-1.45	Stable -	4
Player 8 + Player 14	33.80	3	-0.98	0.17	-1.14	Stable -	8
Player 4 + Player 8	17.55	3	-0.74	0.17	-0.91	Stable -	5
Player 8 + Player 10	46.82	3	-0.66	0.17	-0.83	Stable -	8
Player 1 + Player 13	26.08	5	-0.50	0.17	-0.66	Stable -	8

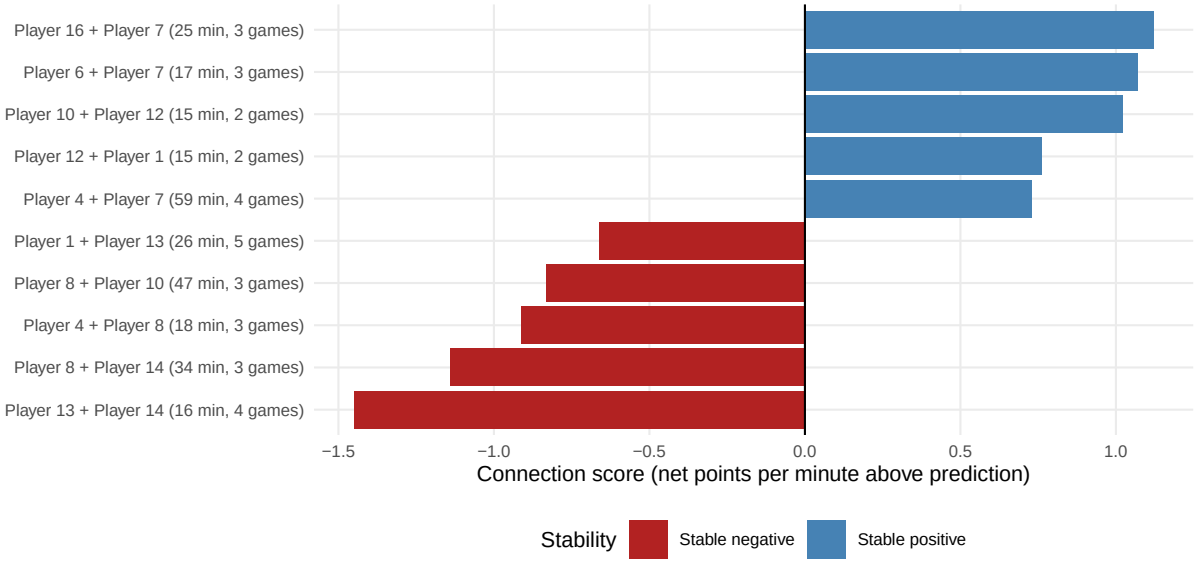


Figure 1: *Connection scores for the five highest and five lowest pairs. Bars are net points per minute above model prediction; colour shows leave-one-game-out stability. Pairs share at least 15 minutes across at least two games.*

Sensitivity of the pair scores

The number of eligible pairs was 85 at ten shared minutes, 73 at fifteen and 56 at twenty (Table 3). Under `lambda.1se`, the scores of pairs eligible at more than one threshold were unchanged by construction.

Using `lambda.min` instead of `lambda.1se`, Spearman correlations with the main scores ranged from 0.998 to 0.999, and between 96.4% and 97.3% of common pairs kept the same sign. Rankings were therefore closely aligned under the two penalty rules, although a small number of signs changed. This does not establish robustness to incomplete coverage, lineup errors or future games.

Table 3: *Pair-score sensitivity relative to the main specification: lambda.1se, 15 shared minutes and at least two games. Agreement is calculated among common eligible pairs.*

Penalty	Minimum minutes	Pairs	Common pairs	Spearman	Same sign (%)
lambda.1se	10	85	73	1.000	100.000
lambda.1se	15	73	73	1.000	100.000
lambda.1se	20	56	56	1.000	100.000
lambda.min	10	85	73	0.999	97.260
lambda.min	15	73	73	0.999	97.260
lambda.min	20	56	56	0.998	96.429

Connections within individual games

For Figure 2, the same season-model residuals are averaged within each pair and game. These scores are descriptive and in-sample. They do not use the held-out prediction procedure. A two-minute within-game display threshold is used; this is not a reliability guarantee. Nodes and colours are consistent across panels, and a missing edge indicates insufficient displayed exposure, not a score of zero. Game panel numbers follow fixture-ID order and are not necessarily chronological.

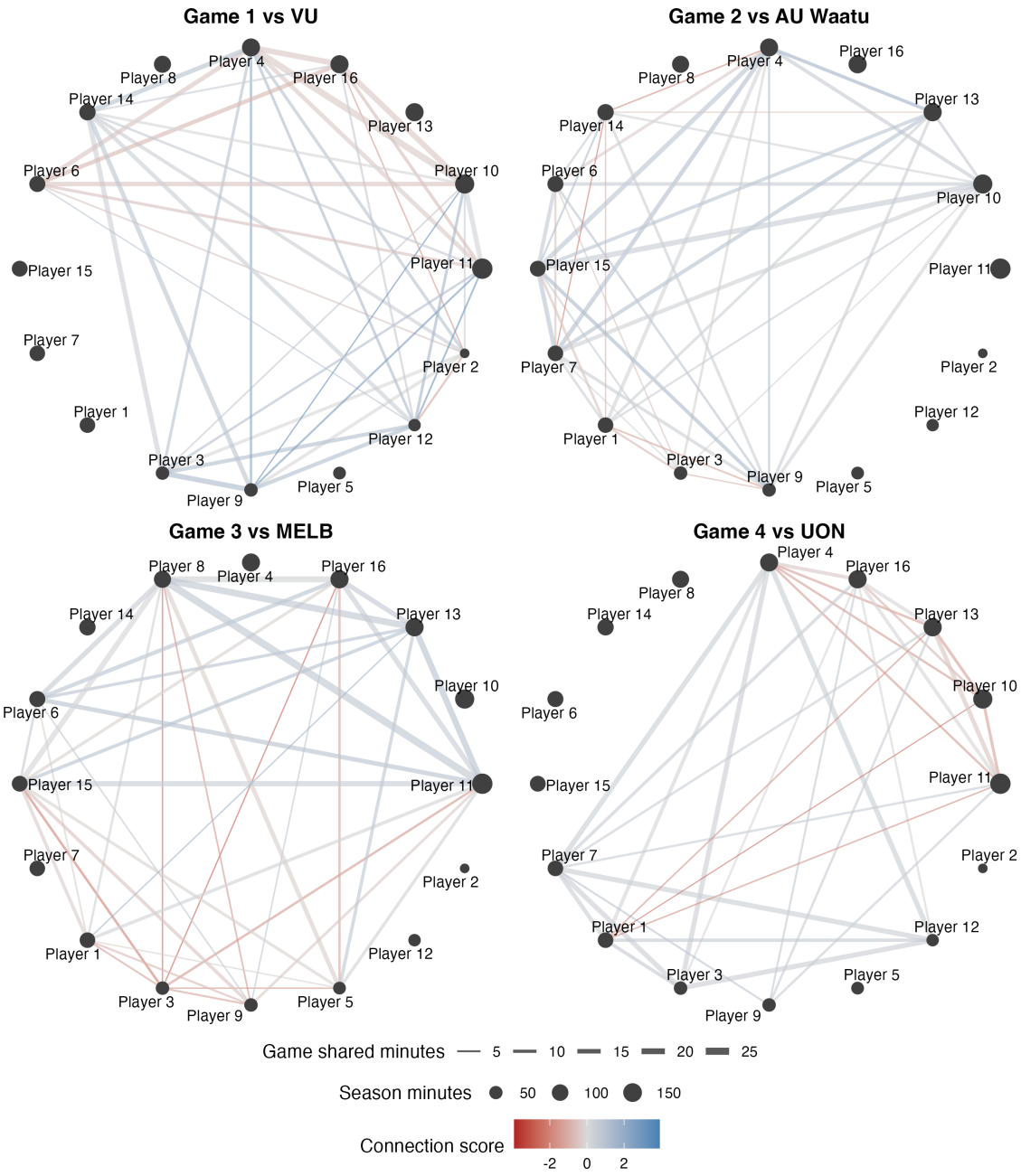


Figure 2: *Pair connection scores for Games 1–4. Node positions and scales are shared with the following figure. Labels identify players, node size represents season minutes, and edges show shared minutes and connection scores.*

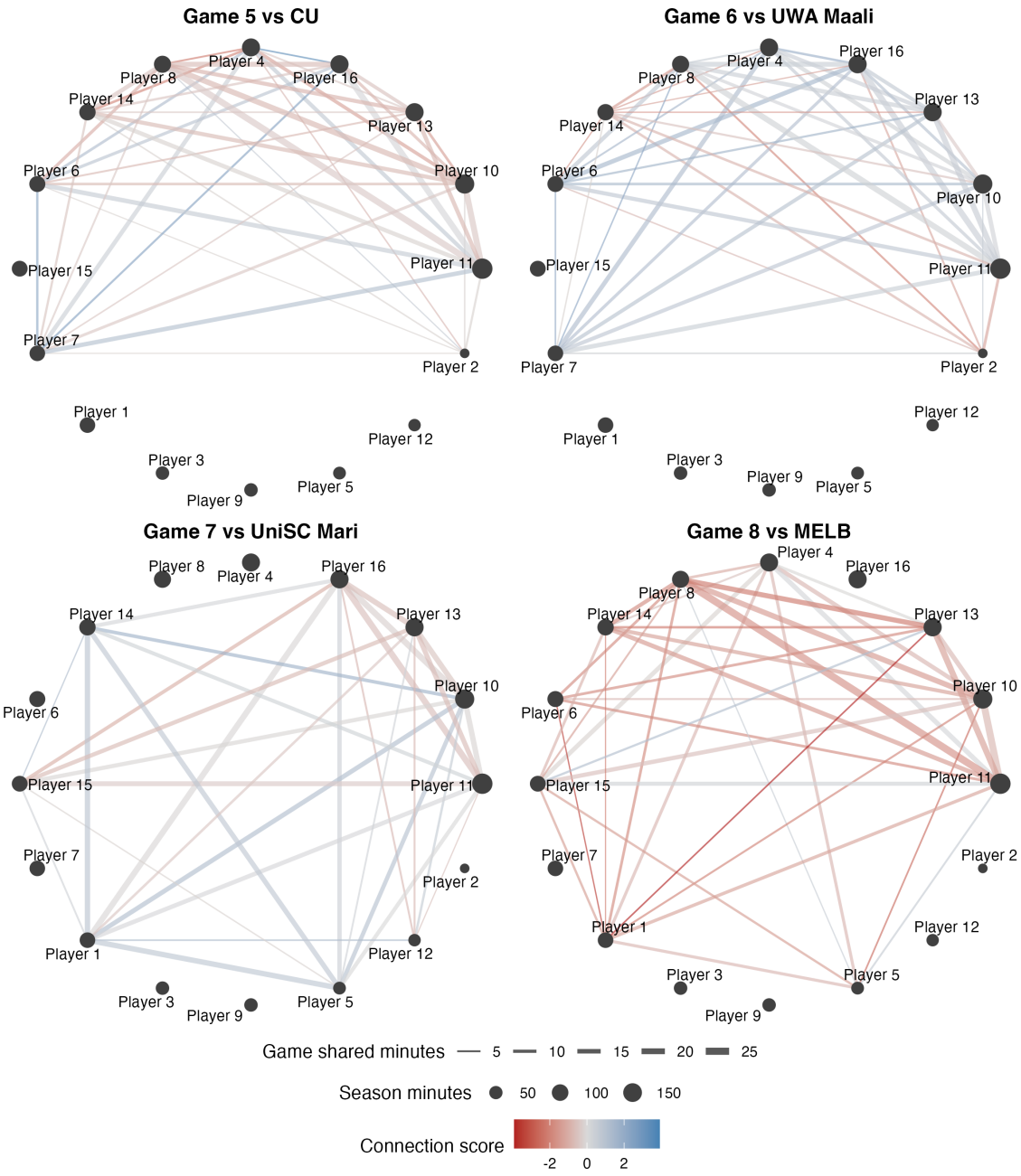


Figure 3: *Pair connection scores for Games 5–8, using the same node positions and scales as Games 1–4.*

Connection network and rotation structure

The connection-score network in Figure 4 contains eligible adjusted pair scores. It differs from the usage network used to calculate centrality: the latter uses a threshold of one shared minute in one game, rather than 15 minutes in two games.

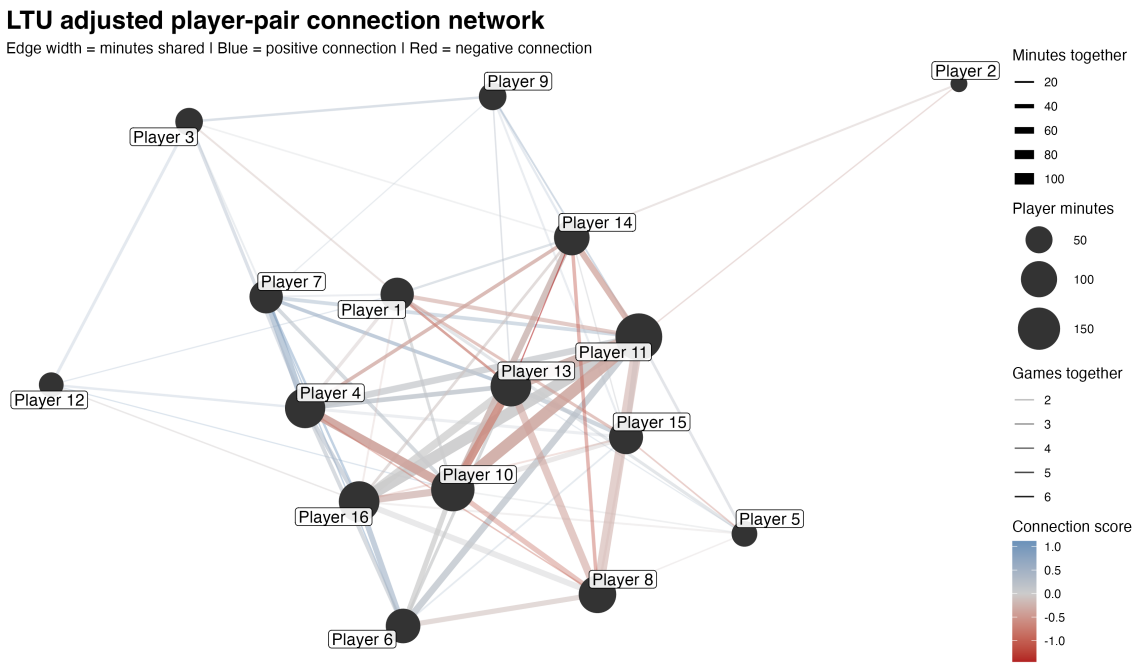


Figure 4: *Network of pairs with a connection score. Nodes are players (size is total minutes), edge width is shared minutes, and edge colour shows the connection score (blue positive, red negative). Players are anonymised.*

In the shared-minute usage network, 112 of 120 possible pairs shared the court (density 0.933; Table 4). Shared-minute exposure was spread fairly evenly (Gini 0.27), whereas weighted betweenness was highly concentrated (Gini 0.88): 11 of 16 players had a betweenness of zero (Figure 5).

The two highest bridging scores belong to Player 11 (0.42) and Player 4 (0.19). They rank 1st and 5th of 16 for shared minutes, so the pattern is consistent with heavy exposure. Betweenness is not simply a function of minutes, however: Player 10, second for shared minutes, has a betweenness of 0.00. The pattern describes deployment and does not establish tactical leadership.

Table 4: *Usage-network summary (pairs with at least one shared minute).*

Measure	Value
Players	16
Connected pairs	112
Possible pairs	120
Density	0.933
Gini of weighted strength	0.267
Gini of weighted betweenness	0.881
Freeman betweenness centralisation (unweighted)	0.002

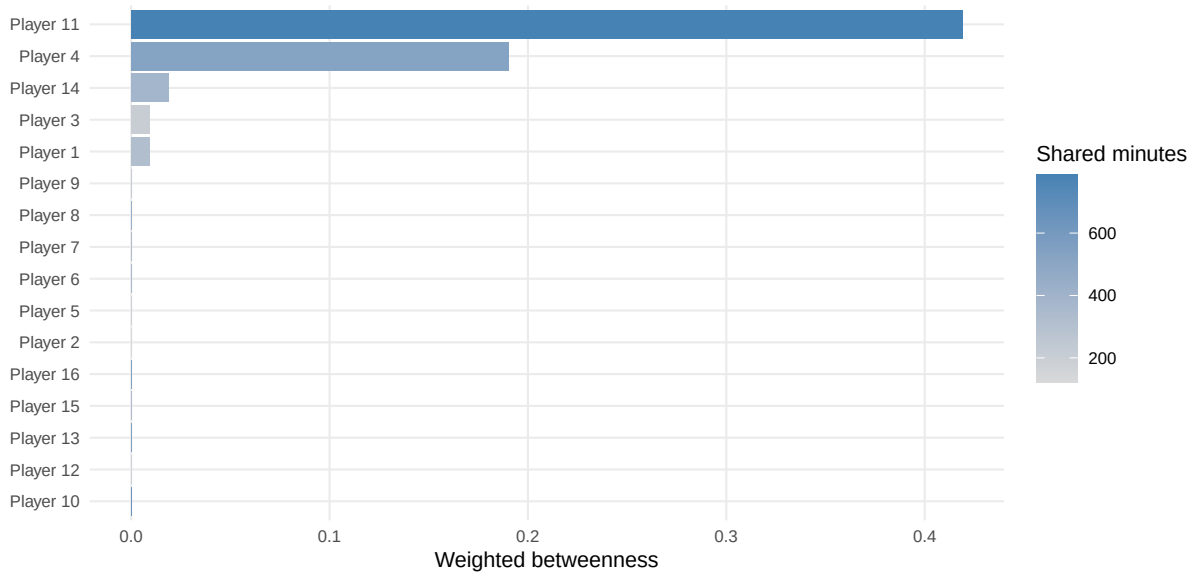


Figure 5: *Weighted betweenness by player in the shared-minute network. Bar shading shows weighted strength (total shared minutes).*

Situational coverage

LTU spent 44% of observed minutes leading by at least six points, 44% within five points and 12% trailing by at least six (Figure 6). Trailing situations appeared in only 3 of the retained games. The clutch window contained 14 stints totalling 9.4 minutes across 3 games. Only 3 lineup-by-situation combinations reached five minutes across two games. This is too little to compare closing lineups, so the situational results are descriptive.

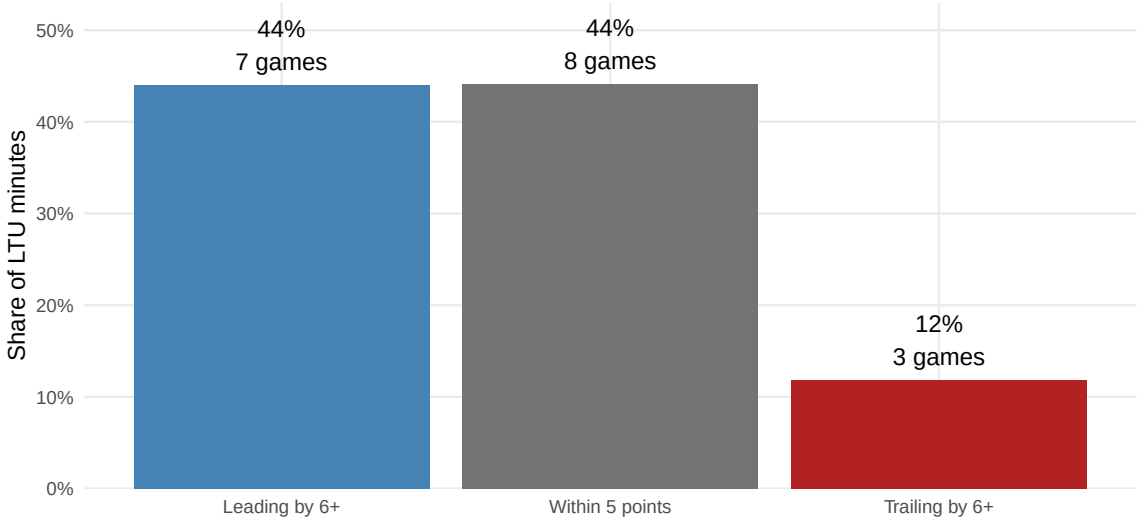


Figure 6: *Share of LTU minutes by score state at stint start, labelled with the number of games in which each state occurred.*

Predictive performance

Across 225 held-out stints, weighted RMSE was 2.303 for the baseline model and 2.300 with the pair adjustment (difference 0.003; positive values favour the adjustment; Table 5). The adjustment lowered error in 5 of 8 held-out games (Figure 7). The weighted correlation between the pair feature and the baseline residuals was 0.054. The overall difference was small, and the exploratory permutation test provided no clear evidence of a predictive advantage (p = 0.219). These findings apply to the tested average-pair adjustment and do not establish that all models of player interactions would fail.

Table 5: *Held-out weighted RMSE by game. Difference is baseline minus pair-adjusted error, so positive values mean the pair scores helped.*

Game	Stints	Baseline	With pair scores	Difference	Improved
Game 1 vs VU	29	2.368	2.419	-0.052	No
Game 2 vs AU Waatu	21	2.442	2.440	0.002	Yes
Game 3 vs MELB	32	2.317	2.375	-0.058	No
Game 4 vs UON	32	2.353	2.332	0.021	Yes
Game 5 vs CU	33	2.219	2.159	0.061	Yes
Game 6 vs UWA Maali	30	1.910	1.876	0.034	Yes
Game 7 vs UniSC Mari	23	2.248	2.272	-0.024	No
Game 8 vs MELB	25	2.545	2.496	0.049	Yes
All	225	2.303	2.300	0.003	5 of 8

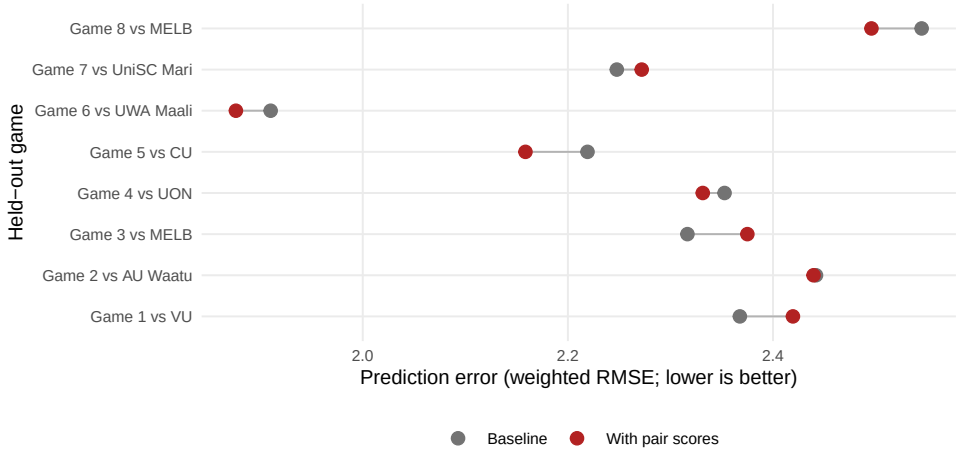


Figure 7: *Held-out prediction error by game. A point to the left of the grey point (lower error) means the pair scores helped in that game.*

6 Discussion

Findings in relation to the objectives

Lineup reconstruction (Objective 1). The reconstruction produced usable stints for 8 of 10 games (Table 1). The coverage and scoring checks show internal consistency in most retained games, but they cannot verify player identities, and the excluded games show that opening-lineup inference can fail. The pair and network results therefore depend on lineups that are inferred rather than observed.

Pair performance (Objective 2). Table 2 and Figure 1 identify combinations whose shared stints were above or below the model's prediction. The highest score belongs to Player 16 and Player 7 (1.12 net points per minute above prediction over 25 shared minutes and 3 games), and Player 7 appears in 3 of the five highest-scoring pairs. These are candidates for review, not demonstrated partnerships: the extremes were selected from 73 pairs, and each stint's residual is credited to ten overlapping pairs. The Spearman correlation between pair scores and the average of the two players' mean residuals was 0.757. Because both summaries use the same stint residuals, this overlap shows how difficult it is to separate player-associated performance from pair-specific interaction; it does not quantify how much of a pair's score is attributable to one player.

Rotation structure (Objective 3). The usage network was almost complete (Table 4), so the rotation was broadly connected, and bridging positions were concentrated in two heavily used players (Figure 5). Because their shared-minute rankings were 1st and 5th, the pattern is consistent with exposure, although betweenness is not simply a function of minutes. Situational coverage was thin (Figure 6): the clutch window contained only 9.4 minutes, so no closing lineup can be recommended.

Validation (Objective 4). The stability labels and the sensitivity checks (Table 3) show how far the descriptive ranking depends on individual games and analyst choices. They do not show repeatability. The held-out test addresses that question directly, and the pair adjustment did not clearly improve prediction (Table 5 and Figure 7). Possible reasons include the small number of games, the noisy per-minute outcome, the fixed-coefficient form of the adjustment and the unavailable opponent term for most held-out games. The data cannot separate these explanations.

League context (Objective 5). The league networks (Appendix B) show that team strength was concentrated in both competitions and that the leading women's team was not the leading men's team. This is descriptive context for one season; it does not validate the player-pair scores.

Implications for coaching practice

For coaches, the most defensible use is targeted video review. The pair table can point to stretches in which to examine spacing, defensive coverage and the contribution of the other

three players. The usage network can describe which players link rotation groups, without assigning tactical leadership from centrality alone. The results give no basis for changing rotations on predictive grounds. The main significance of the work is methodological and practical: it shows how play-by-play data can be turned into reviewable evidence for a small team, and it demonstrates the value of testing such evidence on games it was not estimated from.

Limitations

- **Sample:** one season, one team's perspective, and 8 retained games. Excluded games also remove data, and a coverage screen of 50% is permissive.
- **Lineup reconstruction:** opening lineups are inferred, and lineups are carried between periods without independent verification.
- **Outcome measure:** performance is per minute rather than per possession, so pace differences remain, and short stints give noisy rates.
- **Controls:** the opponent control is team-level and regularised, and most opponents appear in only one retained game. Score state is not in the baseline model.
- **Interpretation:** a pair's residual can reflect the contributions of the other three players or game circumstances. Many overlapping pairs create opportunities for extreme rankings by chance, and residuals are in-sample.
- **Validation:** stability labels reuse overlapping games, the sensitivity checks vary only a few settings, and the permutation benchmark ignores within-game dependence.
- **League analysis:** one season, with margin-weighted results and different schedules across competitions.

Future work

Future work should prioritise verified lineups and archived source data, then additional games and seasons (for example 2026, if available). Useful additional checks include stricter coverage rules, other stint thresholds and the exclusion of partially covered games. A larger dataset could support possession-adjusted outcomes, controls for opposing lineups and a pair adjustment calibrated within each training fold. A temporal split would better assess prediction of future games than leave-one-game-out validation within a season.

7 Conclusion

This project developed a workflow for examining player connections and rotation structure in LTU women's basketball. Of 73 eligible pairs, 35 had stable positive and 28 stable negative connection scores, and Player 7 appeared in 3 of the five highest-scoring pairs. The usage network showed broad player overlap with concentrated bridging positions, and the league networks showed concentrated team strength in both competitions. However, the tested pair-score adjustment did not clearly improve held-out prediction.

The practical recommendation is to use these results to prioritise video review and data collection. The main limitations are the small single-season sample, inferred lineups and per-minute outcomes. More verified games, and evidence that patterns repeat, are needed before using the scores to recommend changes to rotations.

Appendix A: Software and reproducibility

Analyses were run in R (R Core Team 2026) (https://www.R-project.org/) using the tidyverse (Wickham et al. 2019) (https://www.tidyverse.org/), `glmnet` (Friedman, Hastie, and Tibshirani 2010) (https://CRAN.R-project.org/package=glmnet) and `igraph` (Csárdi and Nepusz 2006) (https://igraph.org/r/). Figures used `ggplot2` and `ggraph` (https://CRAN.R-project.org/package=ggraph), and this report was rendered with Quarto (https://quarto.org/).

The runner, `run-analysis.R`, executes four scripts in order: `R/LTU-team.R` loads saved inputs, reconstructs stints and calculates connection scores and situational summaries; `R/LTU-prediction.R` evaluates held-out predictions; `R/LTU-network-by-game.R` produces per-game networks; and `R/LTU-league-comparison.R` calculates league-level centrality scores. The sensitivity and diagnostic tables read by this report are also saved to `outputs/tables/`. Random seeds are set in the runner and the model code.

The report is rendered separately from the repository's `report/` folder. It reads saved outputs and the input snapshot without downloading new data. Report tables and figures use anonymous player labels; `data/DATA.md` describes what the saved inputs contain and the conditions for sharing them. Instructions for rerunning the analysis are in `README.md`; optional refresh instructions are in `data/DATA.md`. An optional interactive version of the connection network (using `visNetwork`) is created by the team script when that package is installed; the report uses static figures.

Appendix B: League context

For context, a team-level network was built from all UBL results in each 2025 competition. Each match adds an arrow from the losing team to the winning team, weighted by winning margin, and eigenvector centrality ranks teams so that wins over highly ranked teams count for more (Keener 1993; Bonacich 1972). Scores are scaled so the top team equals 1, and the Gini coefficient summarises how concentrated these scores are.

The women's network contains 12 teams and the men's network contains 12. In the women's competition UTAS ranked first and LTU ranked 3rd (Gini 0.527). In the men's competition UON ranked first and UTAS ranked 4th (Gini 0.569), so team strength was specific to each competition (Figure 8).

The men's scores were more concentrated than the women's under this network definition. This describes inequality in the calculated centralities; it does not establish which competition was more competitive overall. The comparison incorporates winning margins and the pattern of opponents, and it may also reflect differences in schedules and network connectivity. Separate scaling within each competition means that a women's team and a men's team with the same score do not necessarily have equivalent playing strength. Women's and men's squads also contain different players, so the university name alone does not determine network position across competitions. These rankings provide descriptive league context; they do not establish a causal explanation for team success.

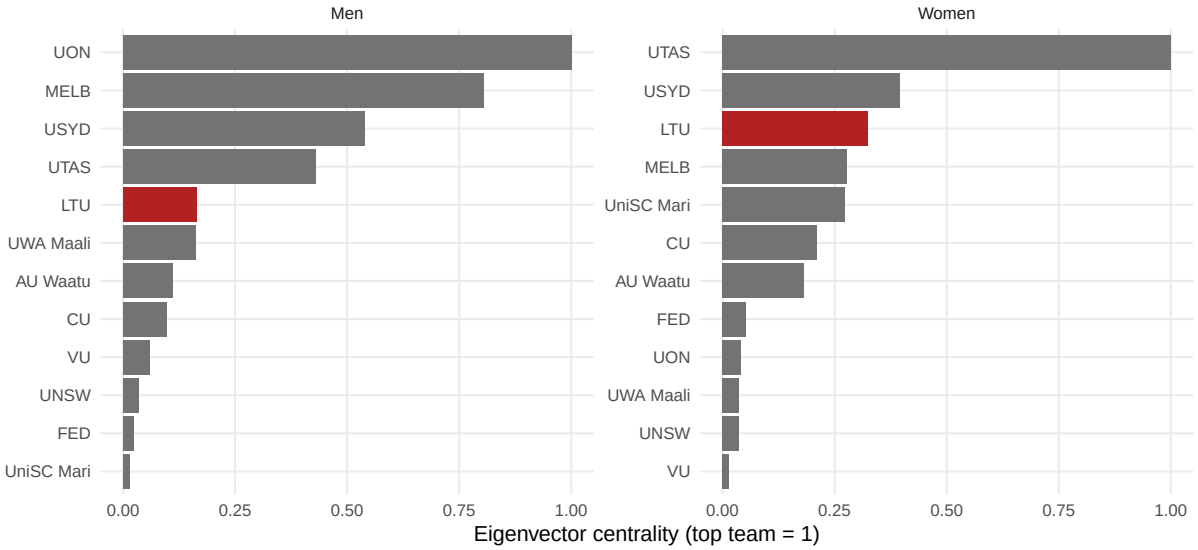


Figure 8: *Eigenvector centrality of teams from the 2025 results network, by competition. LTU is highlighted.*

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AI Acknowledgement

AI tools were used to support the preparation of this report for educational purposes. Their use was limited to assisting with code debugging, wording refinement, and report formatting.

All AI-generated content was critically reviewed, edited, and adapted by the author. The final modelling decisions, analysis, interpretation, and recommendations are my responsi-

bility.